

Power and resource tradeoffs

Project: Reproducible AI Research Study

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Explain the mechanism and terminology of power and resource tradeoffs

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Pilot experiments

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of pilot experiments

Reference example, not the only acceptable wording: Teacher explanation: A pilot checks feasibility, measurement and failure handling before the primary study. Pilot results used for revision are development evidence. Run a small fixture set to test data capture, timing boundaries, rubric and analysis code. Fix defects and freeze the revised plan before new primary cases.

Privacy and data governance

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of privacy and data governance

Reference example, not the only acceptable wording: Teacher explanation: Data governance specifies who may use which records, for what purpose and for how long. Coded data can still be reidentifiable. For fictional records separate participant-code table from analysis values; limit access and retention. Remove unnecessary fields and small-group details before a permitted aggregate export.

Sensitivity analysis

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of sensitivity analysis

Reference example, not the only acceptable wording: Teacher explanation: Sensitivity analysis checks whether a conclusion changes under plausible settings or analysis choices. It should not secretly select the most favourable result. Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range. Settings affecting data roles require special caution.

Two 45-minute sessions. Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the mechanism and terminology of power and resource tradeoffs

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Prerequisite answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

End goal for the pictorial case

Explain power and resource tradeoffs using the records and boundaries of this fictional investigation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term · Meaning

Term: Power

Meaning: The test's detection probability under a specified alternative

Term: Simulation trial

Meaning: A generated dataset under a declared model

Term: Planning effect

Meaning: The effect size used for design calculations

Power and resource tradeoffs — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: For a fixed alternative 100 simulations detect 62 times at $n=20$ and 83 at $n=40$. Per-unit cost 3 credits.

Check or calculation: Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Conclusion and limit: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured. Qualification: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

For a fixed alternative 100 simulations detect 62 times at $n=20$ and 83 at $n=40$. Per-unit cost 3 credits.

Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Power and resource tradeoffs

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

The facts we will use

There are 100 simulation trials per design. At n=20, 60 detect the specified effect; at n=40, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

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Stipulated simulation and resource records

	Independent units	Detections	Trials	Cost units
Smaller design	20	60	100	40
Larger design	40	80	100	80

Read a cell using both its row and column labels.

Power and resource tradeoffs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Estimate simulated detection

Observed detection fractions are $60/100=60\%$ and $80/100=80\%$.



2. Calculate design cost

$n=20$ costs 40 resource units; $n=40$ costs 80 units.



3. Compare the trade-off

Doubling cost adds 20 percentage points of simulated detection under these assumptions.

CHANGE ONE THING

Change the assumed effect or variability but keep the sample sizes.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Estimate simulated detection

What happens and why: Observed detection fractions are $60/100=60\%$ and $80/100=80\%$.

Step: 2. Calculate design cost

What happens and why: $n=20$ costs 40 resource units; $n=40$ costs 80 units.

Step: 3. Compare the trade-off

What happens and why: Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Power is the probability of detecting a specified effect under a specified model and decision rule. Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

What this example does not establish

Only 100 trials per design produce noisy estimates. They are fictional stipulated results, not executed simulation or evidence about pupils.

Change one thing in the pictured case

Change: Change the assumed effect or variability but keep the sample sizes.

Prediction: The detection fractions need new simulation; the resource costs stay 40 and 80.

Reason: Power depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Time or stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Time or stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Time or stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Time or stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Time or stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Time or stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Time or stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Reduce help deliberately

Completion task: For a fixed alternative 100 simulations detect 62 times at $n=20$ and 83 at $n=40$. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is

understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Power and resource tradeoffs. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?

Reasoned teacher answer

The fractions are 60% and 80%; costs are 40 and 80 units. Forty extra units add 20 percentage points here. These simulation estimates do not establish real-study power without justified assumptions.

Changed-case reasoning: The detection fractions need new simulation; the resource costs stay 40 and 80. Power depends on the data-generating assumptions, while this stated unit-cost calculation does not.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

The worked result establishes every future case: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured. Qualification: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision

rule and output.

Teacher explanation: Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget. Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

W2 Trace the worked case

For a fixed alternative 100 simulations detect 62 times at $n=20$ and 83 at $n=40$. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

W3 Audit the limit

A report omits this qualification: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

One hundred alternative trials detect 70 at $n=30$ and 85 at $n=60$; unit cost 2. Compare.

Teacher explanation: 70% versus 85%; cost 60 versus 120. Larger n improves detection here but not automatically under every model.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true. The new case should test this particular mechanism rather than merely restate the original result.

Independent assessment and fresh retry

W4: 70% versus 85%; cost 60 versus 120. Larger n improves detection here but not automatically under every model. Require intermediate reasoning and the stated scope. Fresh retry: Fifty alternative simulations detect 35; sample $n=25$, cost 4 each. Expected reasoning: Estimated detection 70%; study

cost 100 credits. Finite simulation uncertainty remains. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case: One hundred alternative trials detect 70 at $n=30$ and 85 at $n=60$; unit cost 2. Compare.

Selected case answer

Teacher explanation: 70% versus 85%; cost 60 versus 120. Larger n improves detection here but not automatically under every model.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled power and resource tradeoffs evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of power and resource tradeoffs. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this

record to last week's record; a matching answer without a trace is insufficient.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Reproducible environment and records. Preserve the current case record and retrieve its evidence boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-7/24, ai-7/26, ai-7/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.